SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Dr.V.Parthipan

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I LOKESH S (192572151)*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
LOKESH S

Annexure A

Debugging & Optimisation Task

1. A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.
   During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.
2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.
3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS)** network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to

cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.
5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1. Debugging & Optimization Task – WLAN and WWAN Network

Problem Identification

A rapidly growing IT company has deployed a Wireless Local Area Network (WLAN) throughout its six-storey smart office building to support more than 1,200 employees. Employees use laptops, smartphones, IP phones, printers, IoT devices, and cloud-based applications. In addition, field engineers and sales executives use 4G/5G Wireless Wide Area Network (WWAN) services to access company resources from remote locations.
Recently, the company has experienced several networking problems, including slow internet speed during peak hours, frequent Wi-Fi disconnections while moving between floors, weak signal strength in conference rooms, poor video conferencing quality, overloaded wireless access points, unreliable WWAN connectivity in rural areas, and outdated wireless security protocols that expose the network to cyber threats.

Root Cause Analysis

The major causes of these problems are:

- High user density causing wireless congestion.
- Poor placement of Access Points (APs), creating coverage gaps.
- Channel interference due to overlapping Wi-Fi channels.
- Improper roaming configuration between APs.
- Uneven distribution of client devices, causing some APs to become overloaded.
- Weak cellular signal strength in remote locations affecting WWAN connectivity.
- Use of outdated security protocols such as WEP and WPA instead of WPA3.
- Lack of proper Quality of Service (QoS) configuration for video conferencing and VoIP traffic.

Debugging Process

The network engineer should perform the following debugging steps:

1. Monitor network traffic using WLAN monitoring tools.
2. Check bandwidth utilization during peak office hours.
3. Perform a wireless site survey to identify weak signal areas.
4. Measure Received Signal Strength Indicator (RSSI) values.
5. Identify channel interference using Wi-Fi analyzer software.
6. Verify roaming performance while users move between floors.
7. Analyze Access Point utilization to detect overloaded APs.
8. Check WWAN signal quality for field engineers.
9. Perform security auditing to identify devices using outdated encryption.
10. Monitor packet loss, latency, and throughput.

Optimization Techniques

To improve network performance, the following optimizations should be implemented:

- Install additional Access Points in high-density and weak-signal areas.
- Enable Automatic Load Balancing to distribute users evenly across APs.
- Configure Fast Roaming (IEEE 802.11r) to reduce handoff delays.
- Use Dual-Band or Tri-Band Wi-Fi (2.4 GHz, 5 GHz, and 6 GHz).
- Configure non-overlapping wireless channels to reduce interference.
- Upgrade to Wi-Fi 6 or Wi-Fi 6E technology.
- Enable Quality of Service (QoS) to prioritize video conferencing, VoIP, and business applications.
- Upgrade wireless security to WPA3 encryption.
- Implement Network Access Control (NAC) for secure authentication.
- Provide Multi-SIM or SD-WAN connectivity for field engineers to improve WWAN reliability.

- Continuously monitor network performance using centralized management software.

Expected Outcome

After optimization:

- Higher wireless throughput.
- Better signal coverage throughout the office.
- Smooth roaming between floors.
- Stable video conferencing.
- Reduced network congestion.
- Balanced Access Point utilization.
- Improved remote connectivity for field employees.
- Enhanced wireless security.
- Increased employee productivity.

2. Debugging & Optimization Task – ATM Network

Problem Identification

A multinational financial organization uses an Asynchronous Transfer Mode (ATM) network to support online banking, Voice over IP (VoIP), video conferencing, and financial transactions. The ATM network transfers information using fixed-size 53-byte cells and Virtual Circuits (VCs) to guarantee Quality of Service (QoS).
Recently, customers have reported delays in online banking transactions, poor voice quality, video conferencing interruptions, and slow network performance. Monitoring shows congested Virtual Circuits, overloaded ATM switches, and inefficient bandwidth allocation.

Root Cause Analysis

The main causes include:

- Congestion in ATM switches.
- Heavy utilization of Virtual Circuits.
- Improper QoS configuration.
- Incorrect ATM service category selection.
- Poor bandwidth allocation among applications.
- Uneven traffic distribution.
- Insufficient capacity during peak business hours.

Debugging Process

The network engineer should:

1. Monitor ATM switch utilization.
2. Identify congested Virtual Circuits.
3. Measure delay, jitter, and packet loss.
4. Analyze ATM cell loss rate.
5. Verify traffic class configuration.
6. Examine bandwidth allocation.
7. Check switch logs for errors.
8. Test end-to-end network performance.
9. Monitor QoS statistics.
10. Verify traffic shaping policies.

Optimization Techniques

Recommended improvements include:

- Redistribute traffic among available Virtual Circuits.
- Upgrade overloaded ATM switches.
- Configure proper QoS policies.
- Use appropriate ATM service categories:
  - **CBR (Constant Bit Rate)** for Voice
  - **VBR (Variable Bit Rate)** for Video
  - **ABR (Available Bit Rate)** for Data Transfer
  - **UBR (Unspecified Bit Rate)** for Background Traffic

- Apply Traffic Shaping.
- Implement Congestion Control.
- Increase backbone bandwidth.
- Continuously monitor ATM performance.

Expected Outcome

After optimization:

- Faster financial transactions.
- Lower delay and jitter.
- Improved VoIP quality.
- Better video conferencing.
- Efficient bandwidth utilization.
- Higher customer satisfaction.
- Stable ATM network performance.

3. Debugging & Optimization Task – MPLS Network

Problem Identification

A multinational e-commerce company uses an MPLS network to connect headquarters, regional offices, warehouses, and cloud data centers. The network supports ERP, VoIP, cloud applications, inventory management, and customer support.
Employees report slow cloud application access, delayed order processing, poor-quality voice calls, and interrupted video conferencing. Network monitoring reveals congested Label Switched Paths (LSPs), incorrect label mappings, and poor traffic engineering.

Root Cause Analysis

Possible causes include:

- Congested Label Switched Paths.
- Incorrect MPLS label mappings.
- Router configuration errors.
- Uneven bandwidth utilization.
- Lack of MPLS Traffic Engineering.
- Poor QoS implementation.
- Backbone congestion.

Debugging Process

The engineer should:

1. Verify MPLS label tables.
2. Check Label Switched Paths.
3. Monitor router CPU and memory utilization.
4. Measure latency and packet loss.
5. Verify Traffic Engineering configuration.
6. Check routing tables.
7. Monitor bandwidth utilization.
8. Analyze QoS policies.
9. Identify congested backbone links.
10. Perform end-to-end connectivity testing.

Expected Outcome

The optimized MPLS network will provide:

- Faster cloud application access.
- Reduced latency.
- Better VoIP performance.
- High-quality video conferencing.
- Balanced bandwidth utilization.
- Improved order processing.
- Reliable enterprise communication.

4. Debugging & Optimization Task – Error Detection and Correction

Problem Identification

A logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory systems exchange large amounts of data. Recently, shipment records have become inconsistent, barcode scans fail, inventory mismatches occur, and network monitoring reports increased transmission errors.

Root Cause Analysis

The problems are caused by:

- Electrical interference.
- Signal attenuation.
- Heavy network congestion.
- Corrupted data frames.
- Weak error detection methods.
- Inefficient retransmission mechanisms.
- Faulty cables or damaged network devices.

Debugging Process

The engineer should:

1. Monitor transmission error rates.
2. Analyze CRC error reports.
3. Check retransmission statistics.
4. Test physical cables and connectors.
5. Verify barcode scanner communication.
6. Inspect RFID reader performance.
7. Measure packet loss.
8. Monitor network congestion.
9. Analyze database synchronization.
10. Verify error correction mechanisms.

Optimization Techniques

Recommended solutions include:

- Implement Cyclic Redundancy Check (CRC).
- Use Forward Error Correction (FEC).
- Apply Automatic Repeat Request (ARQ).

- Replace faulty network cables.
- Reduce electromagnetic interference.
- Upgrade switches and network interfaces.
- Improve network segmentation.
- Configure congestion control mechanisms.
- Continuously monitor transmission quality.

Expected Outcome

After optimization:

- Accurate shipment records.
- Reliable barcode scanning.
- Correct inventory database.
- Reduced transmission errors.
- Improved data integrity.
- Faster warehouse operations.
- Increased customer satisfaction.

5. Debugging & Optimization Task – Flow Control

Problem Identification

An online education platform serves over 15,000 students simultaneously during live classes and online examinations. During examinations, students experience incomplete downloads, delayed page loading, duplicate packets, retransmissions, receiver buffer overflow, high latency, and reduced throughput.

Root Cause Analysis

The communication problems occur because:

- The server transmits data faster than receivers can process.
- Poor flow control mechanism.
- Receiver buffers become full.
- High network congestion.
- Inefficient retransmission algorithms.
- Bandwidth differences between server and clients.
- Heavy simultaneous access during examinations.

Debugging Process

The network engineer should:

1. Monitor sender transmission rate.
2. Measure receiver buffer utilization.
3. Analyze packet retransmissions.
4. Measure throughput and latency.
5. Monitor bandwidth utilization.
6. Verify TCP flow control configuration.
7. Test congestion control algorithms.
8. Monitor packet loss.
9. Analyze server resource utilization.
10. Simulate peak traffic conditions.

Optimization Techniques

To improve performance:

- Implement the Sliding Window Protocol.
- Configure TCP Flow Control.

- Enable TCP Congestion Control algorithms.
- Increase receiver buffer capacity.

- Prioritize examination traffic using QoS.
- Optimize retransmission timers.
- Deploy Load Balancers across multiple servers.
- Use a Content Delivery Network (CDN) for static resources.
- Upgrade server capacity.
- Continuously monitor network performance.

Expected Outcome

After implementing these optimizations:

- Reliable delivery of question papers.
- Reduced packet loss and retransmissions.
- No receiver buffer overflow.
- Improved throughput.
- Lower latency.
- Better bandwidth utilization.
- Smooth online examinations and live classes.
- Increased reliability and user satisfaction.